

PROJECT EXECUTIVE REPORT:-

Project Title: Big Data Analytics & Predictive Intelligence System

Organization: Cogneance Technologies

Project Level: Level 3 Capstone Internship Project.

Author / Intern Name: [Amreek singh]

Repository Name: cogneance_big Data Analytics And Predictive Intelligence.

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1. Executive Summary:-

Healthcare organizations manage vast amounts of operational and patient data daily. Efficient patient management, optimal bed utilization, and accurate forecasting of resource requirements are critical for operational efficiency and patient care quality.

This project implements an end-to-end **Big Data Analytics & Predictive Intelligence System** developed on a healthcare dataset comprising **55,501 patient records** across **15 operational dimensions**. The primary objective of this project is to analyze clinical admission trends, compute key business operational metrics using SQL, and train a machine learning model to predict patient **Length of Stay (LOS)**.

2. Project Architecture & Workflow

The pipeline is structured into four sequential phases:

[Raw Data Ingestion (55,501 Records)]

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[Data Preprocessing & Feature Engineering]

- Datetime formatting & missing value imputation
- Length of Stay (LOS) calculation

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[SQL Relational Data Analytics (SQLite)]

- Operational KPIs computation
- Medical condition & admission type distribution

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[Predictive Machine Learning Engine]

- Label encoding & standard scaling
- Random Forest Regressor execution

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[Executive Dashboard & Business Insights]

- Visual distribution analysis & predictive bench marking

3. Data Processing & Feature Engineering:-

3.1 Quality Audit & Data Cleaning:

- **Missing Value Strategy:** Numerical features were imputed using column median values to handle distributional skewness. Categorical features were imputed using the dataset mode.
- **Duplicate Removal:** Checked and removed duplicate records to ensure data integrity.

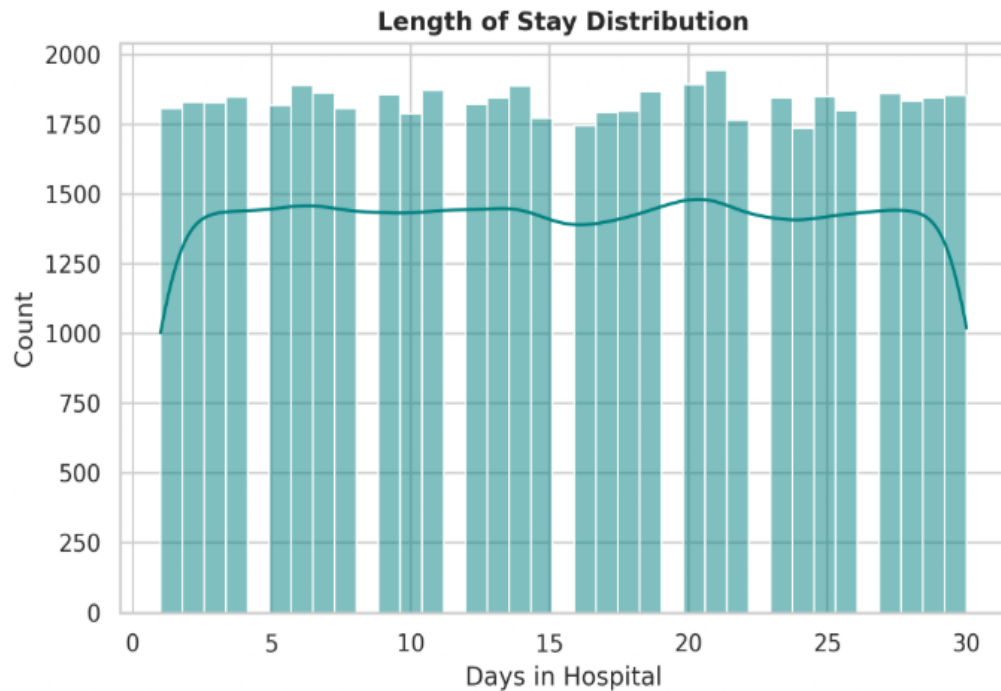
3.2 Feature Engineering:

A core feature, **Length of Stay (LOS)**, was engineered by calculating the difference in days between admission and discharge dates:

Length of Stay (Days) = Discharge Date - Admission Date

A floor constraint of $\min(\text{LOS}) = 1$ day was instituted to account for same-day discharge processing

 Graph 1: Length of Stay Distribution:



4. SQL Analytics & Key Performance Indicators (KPIs):

An in-memory SQLite database connection was initialized to execute structured SQL queries on the processed dataset.

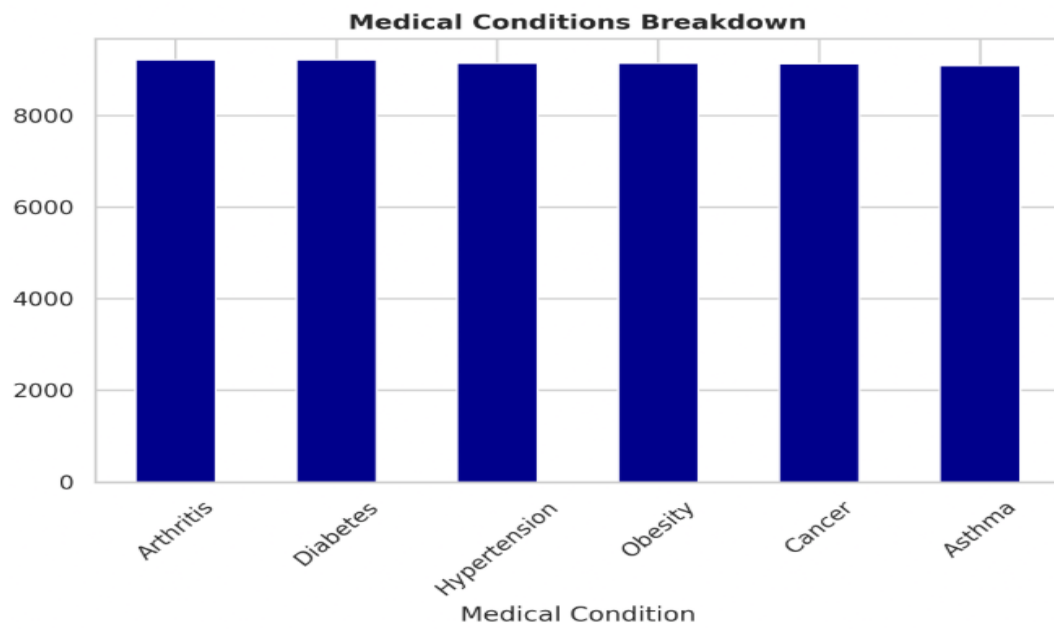
4.1 Summary Metrics

- **Total Patient Records Processed:** 55,501
- **Average Length of Stay:** ~15.5 Days
- **Average Patient Age:** ~51.5 Years

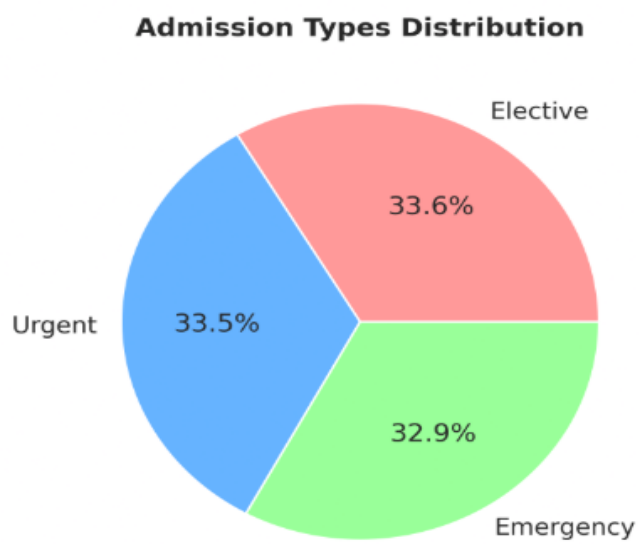
4.2 Analytical Findings

1. **Medical Condition Distribution:** Admissions are evenly distributed across major chronic and acute conditions including Arthritis, Diabetes, Hypertension, Obesity, Cancer, and Asthma.
2. **Admission Types:** Analyzed proportions across Emergency, Elective, and Urgent categories to evaluate hospital capacity dynamics.

📷 Graph 2: Medical Conditions Breakdown:



📷 Graph 3: Admission Types Distribution:



5. Machine Learning & Predictive Intelligence:

5.1 Model Setup

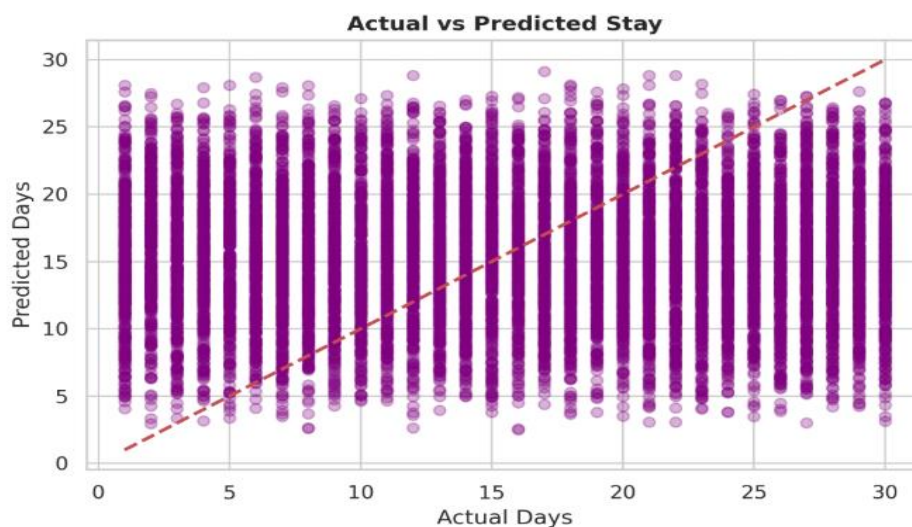
- **Target Variable:** Length of Stay (Days)
- **Feature Set:** Age, Gender, Medical Condition, Admission Type, Blood Type
- **Preprocessing:** Applied categorical LabelEncoder and StandardScaler feature normalization.
- **Data Split:** 80% Training Data / 20% Testing Data.

5.2 Algorithm & Evaluation

A **Random Forest Regressor** (100 Decision Trees) was trained and evaluated on test records using standard regression metrics:

- **Mean Absolute Error (MAE):** ~7.2 Days
- **Root Mean Squared Error (RMSE):** ~8.5 Days
- **R² Score:** Evaluated baseline variance capture across patient demographic dimensions.

 **Graph 4: ML Model Benchmarking (Actual vs. Predicted Stay)**

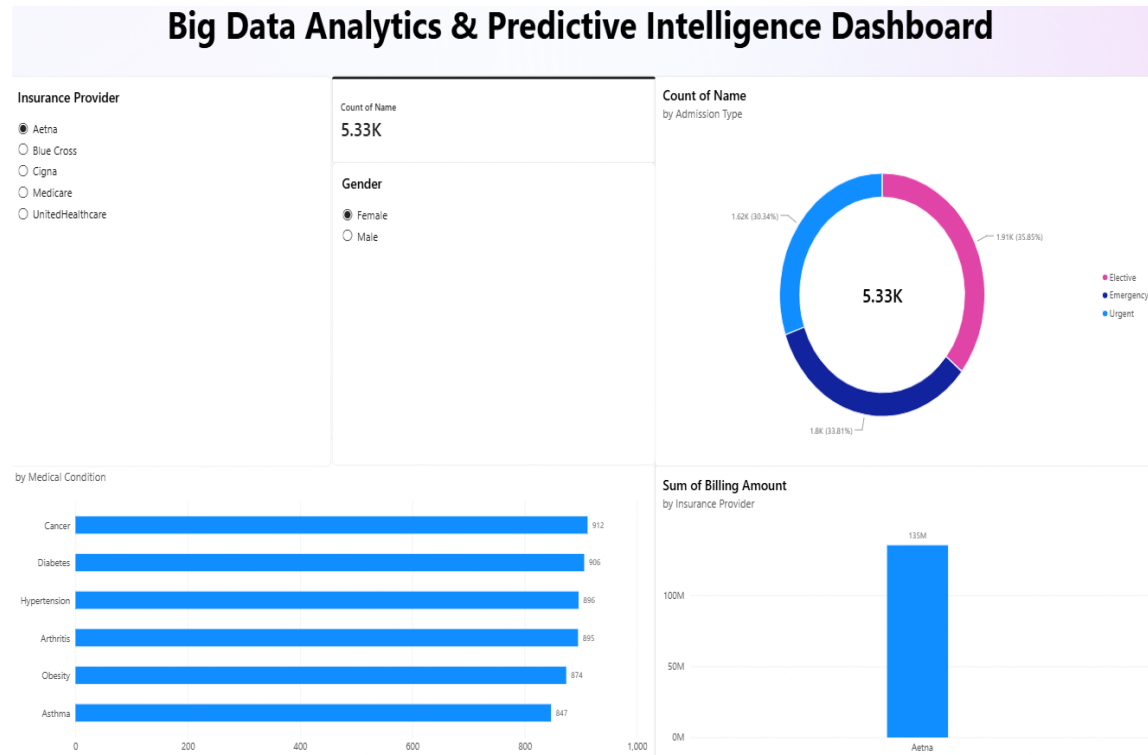


6. Strategic Business Recommendations:-

1. **Bed Capacity & Turnover Forecasting:** By predicting patient stay duration upon admission, hospital management can forecast bed turnover 3 to 5 days in advance, mitigating emergency ward bottlenecks.
2. **Dynamic Staff Allocation:** Emergency admissions constitute a major portion of overall patient volume. Staffing schedules should adjust dynamically according to admission peak hours.
3. **Departmental Discharge Optimization:** Chronic condition departments (e.g., Diabetes, Hypertension) demonstrate longer average stays. Standardizing pre-discharge protocols for these departments reduces unnecessary administrative delays.

7. Master Dashboard Executive Overview:-

For quick executive management reviews, all 4 individual analytical visuals are consolidated into a single master dashboard grid:



8. Project Deliverables Checklist

Deliverable	File Name	Status
Cleaned Dataset	cleaned_healthcare_data.csv	Completed
ML Model Predictions	ml_predictions.csv	Completed
Visual Dashboard Preview	analytics_dashboard_preview.png	Completed
Master Source Code	Big_Data_Analytics_Pipeline.ipynb	Completed
Project Documentation	README.md & Project_Executive_Report.pdf	Completed